MATH 101

Integral Calculus

Detailed Course Summary

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This document covers all 12 weeks of large-class lectures (LC01–LC12) and small-class sessions (SC01–SC12), with key definitions, main ideas, and worked examples (drawn from WeBWorK) for every topic.

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- This document is a **summary of main ideas** — it is **not** a quick-reference cheat sheet. You are expected to have a thorough understanding of all course material, not just what appears here.
- **You cannot pass the course by studying only this document the day before the exam.** There is no shortcut.
- The worked examples included here are questions worth highlighting — they do **not** represent an exhaustive list of exam topics, nor do they guarantee those specific problems will appear on any assessment.
- You are expected to be able to solve **every** WeBWorK problem on your own. If you cannot do a WeBWorK problem independently, do not assume this document covers it — go back and practice it.
- If you are looking for more questions, a good starting point would be the practice problem mentioned on small class worksheet.

Chapter 1 Integration (Weeks 1–7)
Chapter 2 Probability (Weeks 7–9)
Chapter 3 Series and Power Series (Weeks 9–12)

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How to Read This Document

Throughout this summary:

- **Orange boxes** contain a worked *example* drawn from WeBWorK or exam-style problems.
- **Purple boxes** state key *definitions* or theorems.
- **Green boxes** collect tips, tricks, and problem-solving strategies to help you work more efficiently.

Content is organised by week and topic, covering all 12 large-class weeks (LC01–LC12) and the corresponding small-class themes (SC01–SC12).

1 Chapter 1: Integration (Weeks 1–6)

1.1 Week 1 – The Definite Integral (LC01)

Main Idea 1 – The Definite Integral as Area

Definite Integral (geometric)

When $a < b$ and $f(x) \geq 0$ on $[a, b]$,

$$\int_a^b f(x)\,dx = \text{area between the curve } y = f(x) \text{ and the } x\text{-axis on } [a, b].$$

Evaluating by geometry

Evaluate $\displaystyle\int_0^1 \sqrt{1 - x^2}\,dx$.

Solution. The curve $y = \sqrt{1 - x^2}$ is the upper half of the unit circle. The area from $x = 0$ to $x = 1$ is a quarter-circle of radius 1:

$$\int_0^1 \sqrt{1 - x^2}\,dx = \frac{\pi(1)^2}{4} = \frac{\pi}{4}.$$

Main Idea 2 – The Area Function

Area Function

Given an integrable function f and a fixed lower limit a, the *area function* is

$$A(x) = \int_a^x f(t)\,dt.$$

$A(x)$ measures the net signed area accumulated from a to x.

Computing an area function

Let $A(x) = \displaystyle\int_0^x 7\,dt$ for $x \geq 0$. Find $A(3)$.

Solution. The integrand is the horizontal line $f(t) = 7$. The area of the rectangle from 0 to x is $7x$, so $A(x) = 7x$ and $A(3) = 21$.

Main Idea 3 – Right-Endpoint Riemann Sums

Right Riemann Sum

With $\Delta x = (b-a)/n$ and $x_i = a + i\,\Delta x$:

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} f(x_i)\,\Delta x.$$

Right Riemann sum — WW1 P16

Find the left and right Riemann sums for $\displaystyle\int_4^{10} (2x^2+3x+6)\,dx$ with $n=3$.

Solution. $\Delta x = (10-4)/3 = 2$; partition: $4, 6, 8, 10$.
Left (L_3): use $x = 4, 6, 8$:

$$L_3 = 2\big[f(4)+f(6)+f(8)\big] = 2\big[50+60+82+\ldots\big].$$

$f(4) = 2(16)+12+6 = 50, \quad f(6) = 72+18+6 = 96, \quad f(8) = 128+24+6 = 158.$

$$L_3 = 2(50+96+158) = 2(304) = 608.$$

Right (R_3): use $x = 6, 8, 10$: $f(10) = 200+30+6 = 236$.

$$R_3 = 2(96+158+236) = 2(490) = 980.$$

Since f is increasing on $[4,10]$, we have $L_3 < \int_4^{10} f\,dx < R_3$.

Tips & Tricks – Riemann Sums

- If f is **increasing** on $[a,b]$: left sum *underestimates*, right sum *overestimates*. Swap for decreasing f.
- The exact integral is always squeezed between L_n and R_n; increasing n shrinks the gap.
- Sigma notation quick-check: the number of terms equals n, and the last right endpoint is $a + n\Delta x = b$.

1.2 Week 1 Small Class – FTC Part 1 (SC01)

FTC Part 1 application

Let $A(x) = \displaystyle\int_0^x \cos(t)\,dt$. By FTC Part 1, $A'(x) = \cos(x)$.

Verification: Direct integration gives $A(x) = \sin(x)$, and indeed $\frac{d}{dx}\sin(x) = \cos(x)$. ✓

Chain Rule + FTC1 — WW2 P3d

Find $F'(x)$ where $F(x) = \displaystyle\int_{x^2+1}^{1} \frac{1}{t}\,dt$.

Solution. Reverse the limits: $F(x) = -\int_1^{x^2+1} \frac{1}{t}\,dt$. Let $u = x^2+1$; by FTC1 and the Chain Rule:

$$F'(x) = -\frac{1}{u}\cdot\frac{du}{dx} = -\frac{1}{x^2+1}\cdot 2x = -\frac{2x}{x^2+1}.$$

Tips & Tricks – FTC Part 1

- Template: $\displaystyle\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))\cdot g'(x)$. The lower limit a is just a constant and *vanishes* after differentiation.
- If both limits depend on x, split at a constant: $\int_{h(x)}^{g(x)} = \int_c^{g(x)} - \int_c^{h(x)}$, then apply the template to each piece.
- The variable of integration (t) is a *dummy variable* — it disappears in the answer.

1.3 Week 2 – Integral Properties and FTC Part 2 (LC02)

Main Idea 1 – Arithmetic Properties of Integrals

Key Properties

$$\int_a^b Kf(x)\,dx = K\int_a^b f(x)\,dx, \quad K \text{ constant}$$
$$\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$
$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$$
$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

Using additivity

Evaluate $\displaystyle\int_{-1}^{2} |x|\,dx$.

Solution. Split at $x=0$: $|x| = -x$ for $x<0$ and $|x| = x$ for $x \geq 0$.

$$\int_{-1}^{2} |x|\,dx = \int_{-1}^{0}(-x)\,dx + \int_0^2 x\,dx = \left[-\tfrac{x^2}{2}\right]_{-1}^{0} + \left[\tfrac{x^2}{2}\right]_0^2 = \tfrac{1}{2} + 2 = \tfrac{5}{2}.$$

Main Idea 2 – Odd/Even Symmetry

Odd and Even Functions

f is *odd* if $f(-x) = -f(x)$; *even* if $f(-x) = f(x)$. For an odd integrable f: $\int_{-a}^{a} f(x)\,dx = 0$. For an even integrable f: $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$.

Odd function shortcut — WW2 P15d

Evaluate $\displaystyle\int_{-\pi/2}^{\pi/2} \frac{6x^2 \sin x}{1+x^6}\,dx$.

Solution. Let $f(x) = \dfrac{6x^2 \sin x}{1+x^6}$. Check: $f(-x) = \dfrac{6x^2(-\sin x)}{1+x^6} = -f(x)$, so f is odd. Therefore:

$$\int_{-\pi/2}^{\pi/2} f(x)\,dx = 0.$$

Main Idea 3 – FTC Part 2 and Antiderivatives

FTC Part 2

If $F'(x) = f(x)$ on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a) \equiv \Big[F(x)\Big]_a^b.$$

FTC Part 2 — WW2 P10

Evaluate $\displaystyle\int_1^2 \frac{8y - 8y^7}{y^3}\,dy$.

Solution. Simplify: $\dfrac{8y - 8y^7}{y^3} = 8y^{-2} - 8y^4$. Then:

$$\int_1^2 \left(8y^{-2} - 8y^4\right) dy = \left[-8y^{-1} - \tfrac{8}{5}y^5\right]_1^2 = \left(-4 - \tfrac{256}{5}\right) - \left(-8 - \tfrac{8}{5}\right) = 4 - \tfrac{248}{5} = -\tfrac{228}{5}.$$

Main Idea 4 – Indefinite Integrals

Indefinite integral – guess-check-adjust

Find $\displaystyle\int e^{\pi x}\,dx$.

Guess: $F(x) = e^{\pi x}$. *Check:* $F'(x) = \pi e^{\pi x} \neq e^{\pi x}$. *Adjust:* divide by π.

$$\int e^{\pi x}\,dx = \frac{1}{\pi}e^{\pi x} + C.$$

Tips & Tricks – FTC Part 2 & Antiderivatives

- **Always verify** your antiderivative by differentiating it — this takes 10 seconds and catches all errors.
- For $\int f(kx)\,dx$: antidifferentiate f as usual, then *divide* by k. E.g. $\int \sin(3x)\,dx = -\frac{1}{3}\cos(3x) + C$.
- Any constant C in $F(x) + C$ cancels when computing $F(b) - F(a)$, so pick the simplest antiderivative (usually $C = 0$).
- $\int 1/x\,dx = \ln|x| + C$ — the absolute value is required when x can be negative.

1.4 Week 2 Small Class – Integration in Context (SC02)

Net displacement vs. total distance — WW2 P22

A particle has velocity $v(t) = -t^2 + 5t - 6$ m/s. Find the net displacement and total distance over $[0, 6]$.

Net displacement:

$$\int_0^6 (-t^2 + 5t - 6)\,dt = \left[-\tfrac{t^3}{3} + \tfrac{5t^2}{2} - 6t\right]_0^6 = (-72 + 90 - 36) - 0 = -18 \text{ m}.$$

Total distance: Factor $v(t) = -(t-2)(t-3)$; $v < 0$ on $[0, 2]$, $v > 0$ on $[2, 3]$, $v < 0$ on $[3, 6]$.

$$D = \int_0^2 |v|\,dt + \int_2^3 v\,dt + \int_3^6 |v|\,dt = \tfrac{10}{3} + \tfrac{1}{6} + \tfrac{63}{2} = \tfrac{82}{3} \text{ m}.$$

Tips & Tricks – Integration in Context

- **Net displacement** $= \int_a^b v(t)\,dt$ (positive and negative areas cancel).
- **Total distance** $= \int_a^b |v(t)|\,dt$: split the interval at every zero of v and integrate $|v|$ on each sub-interval.
- Units: if v is in m/s and t in s, then $\int v\,dt$ is in metres. Always track units for a sanity check.

1.5 Week 3 – Substitution and Area Between Curves (LC03)

Main Idea 1 – Substitution Rule

Substitution – indefinite integral

Evaluate $\displaystyle\int 2x\,e^{x^2+1}\,dx$.

Solution. Let $u = x^2 + 1$, so $du = 2x\,dx$.

$$\int 2x\,e^{x^2+1}\,dx = \int e^u\,du = e^u + C = e^{x^2+1} + C.$$

Substitution – definite integral, change bounds — WW3 P9

Evaluate $\displaystyle\int_0^1 \frac{24x - 2\pi}{6x^2 - \pi x + \pi}\,dx$.

Solution. Let $u = 6x^2 - \pi x + \pi$, so $du = (12x - \pi)\,dx = \frac{1}{2}(24x - 2\pi)\,dx$. Bounds: $x = 0 \Rightarrow u = \pi$; $x = 1 \Rightarrow u = 6 - \pi + \pi = 6$.

$$\int_0^1 \frac{24x - 2\pi}{6x^2 - \pi x + \pi}\,dx = \int_\pi^6 \frac{2\,du}{u} = 2\ln u\Big|_\pi^6 = 2\ln 6 - 2\ln\pi = 2\ln\tfrac{6}{\pi}.$$

Main Idea 2 – Area Between Curves

Area between curves — WW3 P25

Find the area of the region enclosed by $y = |x|$ and $y = x^2 - 2$.

Solution. Find intersections: $|x| = x^2 - 2$. For $x \geq 0$: $x = x^2 - 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow x = 2$ (taking positive root). By symmetry in x, intersections at $x = \pm 2$. On $[-2, 2]$, $|x| \geq x^2 - 2$ (check $x = 0$: $0 \geq -2$✓).

$$A = \int_{-2}^{2} \big[|x| - (x^2 - 2)\big]\,dx = 2\int_0^2 (x - x^2 + 2)\,dx$$

$$= 2\Big[\tfrac{x^2}{2} - \tfrac{x^3}{3} + 2x\Big]_0^2 = 2\Big[2 - \tfrac{8}{3} + 4\Big] = \tfrac{20}{3}.$$

Tips & Tricks – Substitution & Area Between Curves

Substitution:

- Look for a chunk $g(x)$ whose derivative $g'(x)$ also appears (possibly with a constant factor).
- For definite integrals: change bounds *immediately* when you substitute — this avoids back-substituting at the end.
- Try the denominator or an inner function as your substitution and see if that simplify things .

Area between curves:

- Try to **sketch** the region; a quick graph shows which curve is on top and where they intersect.
- Formula: $A = \int_a^b \big[f_{\text{top}}(x) - f_{\text{bottom}}(x)\big]\,dx$. If the curves cross inside $[a, b]$, split the integral there.
- Integrating in y is often simpler when the curves are given as $x = g(y)$ (e.g. parabolas opening sideways).

1.6 Week 3 Small Class – Trigonometric Integrals (SC03)

Odd power – save one cosine — WW4 P2

Evaluate $\displaystyle\int_0^{\pi/2} \sin^{20} x \cos^5 x \, dx$.

Solution. Save one $\cos x$; write $\cos^4 x = (1 - \sin^2 x)^2$. Let $u = \sin x$, $du = \cos x \, dx$; bounds $0 \to 1$:

$$\int_0^1 u^{20}(1-u^2)^2 \, du = \int_0^1 \left(u^{20} - 2u^{22} + u^{24}\right) du = \frac{1}{21} - \frac{2}{23} + \frac{1}{25} = \frac{2}{10626 - \cdots} = \frac{2}{12075}.$$

Exact: $\dfrac{1}{21} - \dfrac{2}{23} + \dfrac{1}{25} = \dfrac{575 - 1050 + 483}{12075} = \dfrac{8}{12075}$.

Even powers – half-angle formula — WW4 P5

Evaluate $I = \displaystyle\int \cos^4(4x) \, dx$.

Solution. Apply $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$ twice: $\cos^4(4x) = \left[\frac{1}{2}(1 + \cos 8x)\right]^2 = \frac{1}{4}(1 + 2\cos 8x + \cos^2 8x)$. Apply again to $\cos^2 8x = \frac{1}{2}(1 + \cos 16x)$:

$$I = \int \left[\tfrac{1}{4} + \tfrac{1}{2}\cos 8x + \tfrac{1}{8}(1 + \cos 16x)\right] dx = \tfrac{3}{8}x + \tfrac{1}{16}\sin 8x + \tfrac{1}{128}\sin 16x + C.$$

Tips & Tricks – Trigonometric Integrals

- **At least one odd power of $\cos x$:** convert all even powers of $\cos x$ using $\cos^2 x = 1 - \sin^2 x$, then let $u = \sin x$ (the even-power function). The leftover single $\cos x$ becomes du.
  E.g. $\int \sin^{10} x \cos^3 x \, dx$: write $\cos^3 x = (1 - \sin^2 x)\cos x$, then let $u = \sin x$.
- **At least one odd power of $\sin x$:** convert all even powers of $\sin x$ using $\sin^2 x = 1 - \cos^2 x$, then let $u = \cos x$ (the even-power function). The leftover single $\sin x$ becomes $-du$.
  E.g. $\int \sin^3 x \cos^2 x \, dx$: write $\sin^3 x = (1 - \cos^2 x)\sin x$, then let $u = \cos x$.
- **Both powers even:** use $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ and $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ repeatedly until no even powers remain.
- **Mixed trig functions** (tan, sec, cot, csc): convert everything to sin and cos first, then apply the odd/even strategy above.
- Useful identity: $\sin(2x) = 2\sin x \cos x$ (helps simplify products after half-angle substitution).

1.7 Week 4 – Trigonometric Substitution and Partial Fractions (LC04)

Main Idea 1 – Trig Substitution

Trig sub with $\sqrt{a^2 - x^2}$

Evaluate $\displaystyle\int \frac{dx}{\sqrt{1-x^2}}$.

Solution. Let $x = \sin\theta$, $dx = \cos\theta\, d\theta$:

$$\int \frac{\cos\theta\, d\theta}{\sqrt{1-\sin^2\theta}} = \int \frac{\cos\theta}{\cos\theta}\, d\theta = \theta + C = \arcsin(x) + C.$$

Trig sub with $(a^2 + x^2)^2$ — WW4 P10

Evaluate $\displaystyle\int \frac{dx}{(4+x^2)^2}$.

Solution. The denominator has the form $(a^2 + x^2)^2$ with $a = 2$, so let $x = 2\tan\theta$, $dx = 2\sec^2\theta\, d\theta$, $4 + x^2 = 4\sec^2\theta$:

$$\int \frac{2\sec^2\theta\, d\theta}{(4\sec^2\theta)^2} = \int \frac{2\sec^2\theta}{16\sec^4\theta}\, d\theta = \frac{1}{8}\int \cos^2\theta\, d\theta = \frac{1}{8}\cdot\frac{1}{2}(\theta + \sin\theta\cos\theta) + C.$$

Convert back using the right triangle ($x = 2\tan\theta$ gives $\sin\theta = \dfrac{x}{\sqrt{4+x^2}}$, $\cos\theta = \dfrac{2}{\sqrt{4+x^2}}$, $\theta = \arctan\frac{x}{2}$):

$$= \frac{1}{16}\arctan\frac{x}{2} + \frac{1}{16}\cdot\frac{x}{\sqrt{4+x^2}}\cdot\frac{2}{\sqrt{4+x^2}} + C = \frac{1}{16}\arctan\frac{x}{2} + \frac{x}{8(4+x^2)} + C.$$

Trig sub after completing the square — WW4 P13

Evaluate $\int \sqrt{28x - x^2}\, dx$.

Step 1 — Complete the square. Write $28x - x^2 = -(x^2 - 28x)$. Add and subtract $(14)^2 = 196$:

$$28x - x^2 = 196 - (x - 14)^2.$$

So the integral becomes $\int \sqrt{196 - (x-14)^2}\, dx$.

Step 2 — Trig substitution. This has the form $\sqrt{a^2 - u^2}$ with $a = 14$, $u = x - 14$. Let $u = 14 \sin\theta$, so $du = 14\cos\theta\, d\theta$ and $\sqrt{196 - u^2} = 14\cos\theta$:

$$\int 14\cos\theta \cdot 14\cos\theta\, d\theta = 196 \int \cos^2\theta\, d\theta = 98(\theta + \sin\theta\cos\theta) + C.$$

Step 3 — Back-substitute. From $u = x - 14 = 14\sin\theta$: $\sin\theta = \dfrac{x-14}{14}$, $\cos\theta = \dfrac{\sqrt{196 - (x-14)^2}}{14} = \dfrac{\sqrt{28x - x^2}}{14}$, $\theta = \arcsin\dfrac{x-14}{14}$.

$$= 98 \arcsin\frac{x-14}{14} + 98 \cdot \frac{x-14}{14} \cdot \frac{\sqrt{28x - x^2}}{14} + C.$$

$$= 98 \arcsin\frac{x-14}{14} + \frac{(x-14)\sqrt{28x - x^2}}{2} + C.$$

Tips & Tricks – Trigonometric Substitution

- **Which substitution?**
  - $\sqrt{a^2 - x^2}$ or $(a^2 - x^2)^n \quad \Rightarrow \quad x = a\sin\theta$
  - $\sqrt{a^2 + x^2}$ or $(a^2 + x^2)^n \quad \Rightarrow \quad x = a\tan\theta$
  - $\sqrt{x^2 - a^2}$ or $(x^2 - a^2)^n \quad \Rightarrow \quad x = a\sec\theta$
- **Completing the square first:** if the quadratic is $ax^2 + bx + c$, complete the square to reveal one of the three forms above.
- **Back-substitution trick:** draw the right triangle once, label all three sides, then read off $\sin\theta$, $\cos\theta$, $\tan\theta$ directly — no trig identities needed.
- In the context of this course, $\sqrt{\cos^2\theta} = \cos\theta$ (we assume θ is in the principal range where $\cos\theta \geq 0$).

Main Idea 2 – Partial Fractions

Partial fractions — WW4 P23

Evaluate $\displaystyle\int \frac{-9x^2+6}{x^2-2x-8}\,dx$.

Solution. Since degrees are equal, divide first: $\dfrac{-9x^2+6}{x^2-2x-8} = -9 + \dfrac{18x-66}{x^2-2x-8}$. Factor denominator: $(x-4)(x+2)$. Decompose:

$$\frac{18x-66}{(x-4)(x+2)} = \frac{A}{x-4} + \frac{B}{x+2}.$$

$x=4$: $A = (72-66)/6 = 1$. $x=-2$: $B = (-36-66)/(-6) = 17$.

$$\int\left(-9 + \frac{1}{x-4} + \frac{17}{x+2}\right)dx = -9x + \ln|x-4| + 17\ln|x+2| + C.$$

Tips & Tricks – Partial Fractions

- **Check degrees first:** if deg(numerator) $\geq$ deg(denominator), do polynomial long division *before* decomposing. A quick shortcut when the degrees are equal: subtract/add multiples of the denominator to the numerator.
- **Cover-up method** for distinct linear factors: to find A in $A/(x-r)$, multiply both sides by $(x-r)$ and then set $x=r$.
- After finding coefficients, **verify** by recombining over a common denominator.
- Result: $\int A/(x-r)\,dx = A\ln|x-r| + C$. Don't forget the absolute value.

1.8 Week 4 Small Class – Volumes (SC04)

Volume of a cone

A cone of height H and base radius R, point down. At height h, the radius is $r(h) = Rh/H$.

$$V = \int_0^H \pi r^2(h)\,dh = \pi\int_0^H \frac{R^2h^2}{H^2}\,dh = \frac{\pi R^2}{H^2}\cdot\frac{H^3}{3} = \frac{1}{3}\pi R^2 H.$$

Solid of revolution – disk method — WW5 P3

Find the volume of the solid obtained by rotating $f(x) = \dfrac{2}{x+1}$ over $[0,3]$ about the x-axis.

Solution.

$$V = \pi\int_0^3\left(\frac{2}{x+1}\right)^2 dx = 4\pi\int_0^3 \frac{dx}{(x+1)^2} = 4\pi\Big[-\tfrac{1}{x+1}\Big]_0^3 = 4\pi\cdot\tfrac{3}{4} = 3\pi.$$

Tips & Tricks – Volumes of Solids

- **Always draw a picture** — sketch the curve, the solid, and a representative slice before writing the integral.
- **Disk method** (rotate $y = f(x)$ around the x-axis): $V = \pi \int_a^b [f(x)]^2 \, dx$.
- **Washer method** (region between two curves): $V = \pi \int_a^b \left[r_{\text{outer}}^2 - r_{\text{inner}}^2\right] dx$.
- If rotating around $y = k$ instead of the x-axis, shift: $r(x) = f(x) - k$ (or $k - f(x)$ if $k > f$).
- **General cross-sections**: $V = \int_a^b A(x) \, dx$ where $A(x)$ is the cross-sectional area at position x.

1.9 Week 5 – Integration by Parts (LC05)

IBP – repeated application — WW5 P15a

Evaluate $\displaystyle\int_0^1 x^2 e^x \, dx$.

Solution. Apply IBP twice. First: $u = x^2$, $dv = e^x dx$:

$$\int_0^1 x^2 e^x \, dx = \Big[x^2 e^x\Big]_0^1 - 2\int_0^1 x e^x \, dx = e - 2\int_0^1 x e^x \, dx.$$

Second: $u = x$, $dv = e^x dx$: $\int_0^1 x e^x \, dx = \Big[x e^x\Big]_0^1 - \int_0^1 e^x \, dx = e - (e-1) = 1$.

Therefore $\displaystyle\int_0^1 x^2 e^x \, dx = e - 2(1) = e - 2$.

IBP – logarithm (definite) — WW5 P8

Evaluate $I = \displaystyle\int_4^7 \log(3x) \, dx$.

Solution. Let $u = \log(3x)$, $dv = dx$; then $du = \frac{1}{x} dx$, $v = x$.

$$I = \Big[x \log(3x)\Big]_4^7 - \int_4^7 1 \, dx = 7 \log 21 - 4 \log 12 - 3.$$

Tips & Tricks – Integration by Parts (LIATE Rule)

Choose u to be the factor that appears **earliest** in the LIATE list; assign the rest to dv:

$$\underbrace{\mathbf{L}}_{\text{ln,log}} \quad \underbrace{\mathbf{I}}_{\text{arcsin,arctan,...}} \quad \underbrace{\mathbf{A}}_{\text{algebraic: } x^n} \quad \underbrace{\mathbf{T}}_{\text{sin,cos,tan,...}} \quad \underbrace{\mathbf{E}}_{e^x,\, a^x}$$

- **Why LIATE?** Logarithms and inverse trig functions have simple derivatives but no obvious antiderivatives, so they should be u. Exponentials and trig functions are easy to antidifferentiate, so they should be dv.
- **Lone logarithm or inverse trig:** set $dv = dx$ (i.e. $v = x$). E.g. $\int \ln x\, dx$: $u = \ln x$, $dv = dx$.
- **Repeated IBP:** if applying IBP once leaves a similar integral, apply it again (e.g. $\int x^2 e^x\, dx$ needs IBP twice). Keep the same "direction" each time (always let e^x be dv).
- **Cyclic IBP:** if applying IBP twice returns the *original* integral I (e.g. $\int e^x \sin x\, dx$), collect I on one side and solve algebraically.

1.10 Week 5 Small Class – Numerical Integration (SC05)

Right Riemann Sum

Partition $[a, b]$ into n equal subintervals of width $\Delta x = \dfrac{b-a}{n}$ with nodes $x_i = a + i\,\Delta x$. The **Right Riemann Sum** uses the right endpoint of each subinterval:

$$R_n = \Delta x \sum_{i=1}^{n} f(x_i) = \Delta x\big(f(x_1) + f(x_2) + \cdots + f(x_n)\big).$$

Note: $f(x_0) = f(a)$ is *excluded* and $f(x_n) = f(b)$ is *included*.

Trapezoid Rule

Using the same partition, the **Trapezoid Rule** averages the left and right Riemann sums:

$$T_n = \frac{\Delta x}{2}\big(f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\big).$$

All interior coefficients are 2; the two endpoints have coefficient 1.

Error bound: $|E_T| \le \dfrac{M_2(b-a)^3}{12n^2}$, where $M_2 = \max_{[a,b]} |f''(x)|$.

Trapezoid and Simpson error bounds — WW7 P13

Let $f(x) = \cos(x^2)$ with $|f''(x)| \le 4$ and $|f^{(4)}(x)| \le 43$ on $[0,1]$. Approximate $I = \int_0^1 \cos(x^2)\,dx$ with $n = 4$ and bound the errors.

Trapezoidal Rule T_4 ($\Delta x = \frac{1}{4}$, nodes $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$):

$$T_4 = \frac{1/4}{2}\Big(f(0) + 2f(\tfrac{1}{4}) + 2f(\tfrac{1}{2}) + 2f(\tfrac{3}{4}) + f(1)\Big).$$

Error bound: $|I - T_4| \le \dfrac{4 \cdot 1^3}{12 \cdot 4^2} = \dfrac{1}{48}$.

Simpson's Rule S_4:

$$S_4 = \frac{1/4}{3}\Big(f(0) + 4f(\tfrac{1}{4}) + 2f(\tfrac{1}{2}) + 4f(\tfrac{3}{4}) + f(1)\Big).$$

Error bound: $|I - S_4| \le \dfrac{43 \cdot 1^5}{180 \cdot 4^4} = \dfrac{43}{46080}$.

Tips & Tricks – Numerical Integration

- **Coefficient patterns** (all use $\Delta x = (b-a)/n$, nodes $x_0, \dots, x_n$):
  - Right Riemann: $\Delta x \cdot (f_1 + f_2 + \cdots + f_n)$ — endpoints f_0 excluded, f_n included.
  - Trapezoid: $\frac{\Delta x}{2}(f_0 + 2f_1 + \cdots + 2f_{n-1} + f_n)$ — inner coefficients all 2.
  - Simpson: $\frac{\Delta x}{3}(f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 4f_{n-1} + f_n)$ — alternates $4, 2, 4, 2, \dots, 4$; *n must be even.*
- **Accuracy hierarchy**: Simpson $\gg$ Trapezoid $\gg$ Riemann (for the same n).
- To find the minimum n for a given accuracy: isolate n from the error inequality, then round *up* to the next integer (and to the next *even* integer for Simpson).

1.11 Week 6 – Improper Integrals (LC06)

Main Idea 1 – Evaluating Improper Integrals

Improper Integral

An integral is *improper* if (1) one or both limits of integration are $\pm\infty$, or (2) the integrand is unbounded at a point in $[a,b]$. Replace the problematic limit by a parameter and take a limit:

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx,$$
$$\int_a^b f(x)\,dx = \lim_{c\to a^+} \int_c^b f(x)\,dx \quad \text{(if } f \text{ blows up at } a\text{)}.$$

The integral **converges** if the limit is a finite number; otherwise it **diverges**.

Improper integral – both types — WW6 P10

Evaluate $\displaystyle\int_0^1 8\log x\,dx$ and $\displaystyle\int_{-1}^1 \frac{5}{\sqrt{1-x^2}}\,dx$.

(a) The integrand $\to -\infty$ as $x \to 0^+$:

$$\lim_{a\to 0^+}\int_a^1 8\log x\,dx = \lim_{a\to 0^+}\Big[8(x\log x - x)\Big]_a^1 = 8(0-1) - 8\lim_{a\to 0^+}(a\log a - a).$$

Since $\lim_{a\to 0^+} a\log a = 0$, this gives $-8 - 0 = -8$.

(b) Improper at both endpoints:

$$\int_{-1}^1 \frac{5}{\sqrt{1-x^2}}\,dx = 5\Big[\arcsin x\Big]_{-1}^1 = 5\Big(\frac{\pi}{2} - \Big(-\frac{\pi}{2}\Big)\Big) = 5\pi.$$

Main Idea 2 – Comparison Tests

Comparison Tests for Improper Integrals

Let f and g be continuous with $g(x) > 0$ on $[a, \infty)$.

Direct Comparison:

- If $|f(x)| \le g(x)$ and $\int_a^\infty g\,dx$ converges, then $\int_a^\infty f\,dx$ converges.
- If $f(x) \ge g(x) \ge 0$ and $\int_a^\infty g\,dx$ diverges, then $\int_a^\infty f\,dx$ diverges.

Limit Comparison (LCT): If $L = \displaystyle\lim_{x\to\infty}\frac{f(x)}{g(x)}$ exists with $0 < L < \infty$, then $\int_a^\infty f\,dx$ and $\int_a^\infty g\,dx$ either both converge or both diverge.

Comparison test for improper integrals — WW6 P17a

Does $\displaystyle\int_1^\infty \frac{2x^2 - x}{\sqrt{x^5+5}}\,dx$ converge or diverge?

Solution. For large x: $\dfrac{2x^2 - x}{\sqrt{x^5+5}} \approx \dfrac{2x^2}{\sqrt{x^5}} = 2x^{-1/2}$. Apply LCT with $g(x) = x^{-1/2}$:

$$L = \lim_{x\to\infty}\frac{(2x^2 - x)/\sqrt{x^5+5}}{x^{-1/2}} = \lim_{x\to\infty}\frac{(2x^2 - x)\sqrt{x}}{\sqrt{x^5+5}} = \lim_{x\to\infty}\sqrt{\frac{(2x^2-x)^2 x}{x^5+5}} = 2.$$

Since $L = 2 \ne 0$ and $\int_1^\infty x^{-1/2}\,dx$ **diverges** ($p = \frac{1}{2} < 1$), the original integral **diverges**.

Tips & Tricks – Improper Integrals

- **Spot the improperness first:** check for (1) infinite limits and (2) singularities (points where $f \to \pm\infty$) inside or at the boundary of $[a, b]$. Split the integral at every singularity before taking limits.
- **p-test** (memorise these): $\displaystyle\int_1^\infty x^{-p}\,dx$ converges $\Leftrightarrow p > 1$. $\displaystyle\int_0^1 x^{-p}\,dx$ converges $\Leftrightarrow p < 1$.
- **Limit** $\lim_{x\to 0^+} x^p \ln x = 0$ for any $p > 0$ — useful when back-substituting limits involving ln.
- **Comparison test strategy**: identify the dominant term of the integrand for large x (or near the singularity), then compare with the simplest function of the same order.
- If the LCT gives $L = 0$ or $L = \infty$, fall back to the direct comparison test.

1.12 Week 6 Small Class – Simpson's Rule and Error Bounds (SC06)

Simpson's Rule & Error Bounds

With $\Delta x = (b-a)/n$ (n even) and $x_i = a + i\,\Delta x$:

$$\int_a^b f(x)\,dx \approx S_n = \frac{\Delta x}{3}\big(f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+\cdots+4f(x_{n-1})+f(x_n)\big).$$

Error bound: If $|f^{(4)}(x)| \le M_4$ on $[a, b]$, then $|I - S_n| \le \dfrac{M_4(b-a)^5}{180\,n^4}$.

Simpson's Rule

Approximate $\displaystyle\int_0^2 \sin(x^2)\,dx$ with $n = 4$.

Solution. $\Delta x = \frac{1}{2}$; nodes $x_0 = 0$, $x_1 = \frac{1}{2}$, $x_2 = 1$, $x_3 = \frac{3}{2}$, $x_4 = 2$.

$$\begin{aligned} S_4 &= \frac{1/2}{3}\Big[f(0) + 4f\big(\tfrac{1}{2}\big) + 2f(1) + 4f\big(\tfrac{3}{2}\big) + f(2)\Big] \\ &= \frac{1}{6}\Big[\sin(0) + 4\sin\big(\tfrac{1}{4}\big) + 2\sin(1) + 4\sin\big(\tfrac{9}{4}\big) + \sin(4)\Big]. \end{aligned}$$

Simpson's Rule – finding minimum n for a given accuracy — WW7 P10

Find the smallest even integer n so that S_n approximates $I = \int_1^{1.5} \frac{dx}{x}$ with error $\leq 10^{-8}$.

Step 1 – Find $f^{(4)}(x)$. $f(x) = x^{-1}$, so $f' = -x^{-2}$, $f'' = 2x^{-3}$, $f''' = -6x^{-4}$, $f^{(4)}(x) = 24x^{-5}$.

Step 2 – Bound $|f^{(4)}|$ on $[1, 1.5]$. $|f^{(4)}(x)| = 24/x^5$ is *decreasing*, so its maximum occurs at $x = 1$:

$$M_4 = \max_{1 \leq x \leq 1.5} |f^{(4)}(x)| = 24.$$

Step 3 – Apply the error formula. We need $\frac{M_4(b-a)^5}{180\,n^4} \leq 10^{-8}$, i.e.

$$\frac{24 \cdot (1/2)^5}{180\,n^4} \leq 10^{-8} \iff \frac{24}{180 \cdot 32\,n^4} \leq 10^{-8} \iff n^4 \geq \frac{24}{180 \cdot 32 \cdot 10^{-8}} = \frac{24 \times 10^8}{5760}.$$

$$n^4 \geq \frac{10^8}{240} \implies n \geq \left(\frac{10^8}{240}\right)^{1/4}.$$

Step 4 – Round up to the nearest even integer. $(10^8/240)^{1/4} \approx 26.9$, so $n \geq 28$ (next even integer).

Tips & Tricks – Simpson's Rule

- Simpson's rule fits a **parabola** through each pair of sub-intervals (needs 3 points per parabola $\Rightarrow n$ must be even).
- Quick coefficient check: the weights follow the pattern $1, 4, 2, 4, 2, \ldots, 4, 1$; the middle weights alternate 4 and 2.
- Simpson's rule is *exact* for polynomials of degree ≤ 3, which is why it performs so well in practice.

2 Chapter 2: Probability (Weeks 7–9)

2.1 Week 7 – Probability Density Functions (LC07)

Probability Density Function (PDF)

A function $f(x)$ is a **probability density function** for a continuous random variable X if:

(i) $f(x) \geq 0$ for all x, and

(ii) $\int_{-\infty}^{\infty} f(x)\,dx = 1$.

Probabilities are computed as areas: $\Pr(a \leq X \leq b) = \int_a^b f(x)\,dx$.

Note: $\Pr(X = a) = 0$ for any single value a (a point has zero width).

PDF – exponential distribution — WW8 P4

A telephone company models call duration with $p(x) = 0.1e^{-0.1x}$ for $x \geq 0$.
(a) What percentage of calls last between 2 and 3 minutes?
(b) What percentage last 4 minutes or more?

(a)

$$\Pr(2 \leq X \leq 3) = \int_2^3 0.1e^{-0.1x}\,dx = \Big[-e^{-0.1x}\Big]_2^3 = e^{-1/5} - e^{-3/10}.$$

(b)

$$\Pr(X \geq 4) = \int_4^\infty 0.1e^{-0.1x}\,dx = \lim_{b\to\infty}\Big[-e^{-0.1x}\Big]_4^b = e^{-2/5}.$$

Finding constants in a PDF — WW8 P8

A continuous random variable X has PDF

$$f(x) = \begin{cases} ae^x & x < 0, \\ bxe^{-x^2} & x \geq 0, \end{cases}$$

and $\Pr(X \leq 0) = 2\Pr(X \geq 0)$. Find a and b.

Solution. $\Pr(X \leq 0) = \int_{-\infty}^0 ae^x\,dx = a$ and $\Pr(X \geq 0) = \int_0^\infty bxe^{-x^2}\,dx = \frac{b}{2}$. The condition gives $a = 2 \cdot \frac{b}{2} = b$. Using Equation (ii): $a + \frac{b}{2} = 1 \Rightarrow b + \frac{b}{2} = 1 \Rightarrow b = \frac{2}{3}$, hence $a = \frac{2}{3}$.

Tips & Tricks – Probability Density Functions

- **Two mandatory properties:** $f(x) \geq 0$ for all x, and $\int_{-\infty}^\infty f(x)\,dx = 1$.
- $\Pr(X = a) = 0$ for any single point a (a single point has zero width $\Rightarrow$ zero area).
- To find an unknown constant C of some PDF: set $\int_{-\infty}^\infty f(x)\,dx = 1$ and solve. This integral is often improper — handle limits carefully.
- **Sanity check:** $f(x)$ can exceed 1 (it is a *density*, not a probability), but $\int f\,dx$ over any interval must be ≤ 1.
- For piecewise PDFs, split into pieces at the breakpoints and add the integrals.

2.2 Week 7 Small Class – Expected Value (SC07)

Expected Value

The **expected value** (mean) of a continuous random variable X with PDF $f(x)$ is:

$$E(X) = \int_{-\infty}^\infty x\,f(x)\,dx.$$

Geometrically, $E(X)$ is the *balance point* of the distribution. Key property: $E(aX + b) = aE(X) + b$.

Expected value – find C then $E(X)$ — WW8 P11

X has PDF $f(x) = Cxe^{-6x}$ for $x \geq 0$, zero elsewhere. Find C and $E(X)$.

Find C. Using IBP twice: $\int_0^\infty xe^{-6x}\,dx = \frac{1}{36}$. So $C \cdot \frac{1}{36} = 1 \Rightarrow C = 36$.

Find $E(X)$.

$$E(X) = \int_0^\infty x \cdot 36xe^{-6x}\,dx = 36\int_0^\infty x^2e^{-6x}\,dx.$$

Using IBP twice (or $\int_0^\infty x^n e^{-ax}\,dx = n!/a^{n+1}$): $\int_0^\infty x^2e^{-6x}\,dx = \frac{2}{216} = \frac{1}{108}$.

$$E(X) = 36 \cdot \frac{1}{108} = \frac{1}{3}.$$

Tips & Tricks – Expected Value

- Formula: $E(X) = \int_{-\infty}^\infty x\,f(x)\,dx$. This is often an improper integral — write the limit explicitly.
- $E(X)$ is **linear**: $E(aX + b) = aE(X) + b$.

2.3 Week 8 Small Class – Variance and Standard Deviation (SC08)

Variance and Standard Deviation

The **variance** of X measures the average squared distance from the mean $E(X) = E(X)$:

$$\mathrm{Var}(X) = E\big[(X - E(X))^2\big] = \int_{-\infty}^\infty (x - E(X))^2\,f(x)\,dx = E(X^2) - [E(X)]^2.$$

The **standard deviation** $\sigma(X) = \sqrt{\mathrm{Var}(X)}$ is in the same units as X and represents a typical distance of X from its mean.

Variance and standard deviation — WW8 P12 style

X has PDF $f(x) = 2xe^{-x^2}$ for $x \geq 0$ (zero elsewhere). Find $E(X)$, $\mathrm{Var}(X)$, and $\sigma(X)$.

Step 1 – $E(X) = \int_0^\infty 2x^2e^{-x^2}\,dx = \sqrt{\pi}/2$.

Step 2 – $E(X^2) = \int_0^\infty 2x^3e^{-x^2}\,dx$. Let $u = x^2$: $= \int_0^\infty ue^{-u}\,du = 1$.

Step 3 – Using the alternative formula:

$$\mathrm{Var}(X) = E(X^2) - (E(X))^2 = 1 - \frac{\pi}{4}.$$

$$\sigma(X) = \sqrt{1 - \tfrac{\pi}{4}}.$$

Tips & Tricks – Variance & Standard Deviation

- **Two equivalent formulas** — choose whichever is easier to compute:
  - Definition: $\mathrm{Var}(X) = \int_{-\infty}^{\infty} (x - E(X))^2 f(x)\,dx$
  - Shortcut: $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$. Usually faster.
- $\mathrm{Var}(X) \geq 0$ always; $\mathrm{Var}(X) = 0$ only if X is a constant.
- $\sigma(X) = \sqrt{\mathrm{Var}(X)}$ is in the *same units* as X; variance is in units^2.
- **Common mistake:** $E(X^2) \neq [E(X)]^2$.

3 Chapter 3: Series and Power Series (Weeks 9–12)

3.1 Week 9 – Sequences, Series, Divergence Test, Integral Test (LC09)

Sequences and Series

A **sequence** $\{a_n\}$ is an ordered list of numbers. It **converges** to L if $\lim_{n\to\infty} a_n = L$ (a finite number); otherwise it **diverges**.

A **series** $\displaystyle\sum_{n=1}^{\infty} a_n$ is defined as the limit of its partial sums $S_N = \sum_{n=1}^{N} a_n$:

$$\sum_{n=1}^{\infty} a_n = \lim_{N\to\infty} S_N.$$

The series **converges** if this limit is a finite number; otherwise it **diverges**.

Divergence Test & Integral Test

Divergence Test: If $\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ diverges. (If $a_n \to 0$, the test is inconclusive.)

Integral Test: Let f be continuous, positive, and decreasing on $[N_0, \infty)$ with $f(n) = a_n$. Then $\sum_{n=1}^{\infty} a_n$ and $\int_{N_0}^{\infty} f(x)\,dx$ either both converge or both diverge.

p-series: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.

Divergence test

Does $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n+1}$ converge or diverge?

Solution. $\lim_{n\to\infty} \frac{n}{n+1} = 1 \neq 0$. By the Divergence Test, the series diverges.

Integral test — WW9

Does $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}$ converge or diverge?

Solution. Take $f(x) = \dfrac{1}{x(\log x)^2}$: continuous, positive, decreasing on $[2, \infty)$.

$$\int_2^{\infty} \frac{dx}{x(\ln x)^2}.$$

Let $u = \ln x$, $du = dx/x$:

$$\int_{\ln 2}^{\infty} u^{-2}\, du = \Big[-u^{-1} \Big]_{\ln 2}^{\infty} = \frac{1}{\ln 2} < \infty.$$

The integral converges, so by the Integral Test the series **converges**.

p-test

For what p does $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

Solution. By the Integral Test: $\int_1^{\infty} x^{-p}\, dx$ converges iff $p > 1$. Therefore $\sum 1/n^p$ converges iff $p > 1$.

Tips & Tricks – Sequences, Divergence Test & Integral Test

- **Sequence vs. series:** a sequence $\{a_n\}$ is a list; a series $\sum a_n$ is the *limit of its partial sums* $S_N = \sum_{n=1}^{N} a_n$.
- **Divergence test** (quick first check): if $\lim_{n\to\infty} a_n \neq 0$, the series *diverges immediately*. If $a_n \to 0$, the test is **inconclusive** — you must use another test.
- **p-series** reference: $\sum 1/n$ diverges; $\sum 1/n^2$ converges; $\sum 1/\sqrt{n}$ diverges; $\sum 1/n^{1.001}$ converges.
- **Integral test conditions:** f must be continuous, positive, and *decreasing* on $[N_0, \infty)$. If f is not monotone (e.g. oscillates), the integral test *cannot* be applied.
- The integral test shows convergence/divergence but **does not** give the sum. Use it only to classify.

3.2 Week 9 Small Class – Geometric Series (SC09)

Geometric Series

A series $\sum_{n=0}^{\infty} ar^n$ is **geometric** with first term a and common ratio r (constant ratio between consecutive terms). It:

$$\text{converges to } \frac{a}{1-r} \quad \text{if } |r| < 1, \qquad \text{diverges} \quad \text{if } |r| \geq 1.$$

Partial sum formula: $\sum_{n=0}^{N} ar^n = a\,\frac{1-r^{N+1}}{1-r}$ for $r \neq 1$.

Geometric series – find parameter for convergence — WW10 P7

Find c such that $\sum_{n=1}^{\infty} \frac{9}{(1+c)^n} = \frac{1}{8}$.

Solution. This is a geometric series with first term $a = \frac{9}{1+c}$ and ratio $r = \frac{1}{1+c}$. For $|r| < 1$ it sums to $\frac{a}{1-r}$:

$$\frac{9/(1+c)}{1-1/(1+c)} = \frac{9/(1+c)}{c/(1+c)} = \frac{9}{c} = \frac{1}{8}.$$

Solving: $c = 72$.

Evaluating series using geometric series — WW10 P4

Evaluate $\sum_{n=1}^{\infty} \frac{10^n + 3^n}{11^n}$.

Solution. Split into two geometric series:

$$\sum_{n=1}^{\infty}\left(\frac{10}{11}\right)^n + \sum_{n=1}^{\infty}\left(\frac{3}{11}\right)^n = \frac{10/11}{1-10/11} + \frac{3/11}{1-3/11} = \frac{10/11}{1/11} + \frac{3/11}{8/11} = 10 + \frac{3}{8} = \frac{83}{8}.$$

Tips & Tricks – Geometric Series

- **Identify the ratio** r by computing a_{n+1}/a_n. If this ratio is a constant (independent of n), the series is geometric.
- **Formula:** $\displaystyle\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ for $|r| < 1$ (diverges for $|r| \geq 1$).
- **Index shift trick:** if the sum starts at $n = k$ (not $n = 0$), factor out r^k: $\sum_{n=k}^{\infty} ar^n = ar^k \cdot \frac{1}{1-r}$.
- **Partial geometric sum:** $\sum_{n=0}^{N} r^n = \frac{1-r^{N+1}}{1-r}$ (useful for finite sums or error estimates).

3.3 Week 10 – Comparison Tests for Series (LC10)

Direct Comparison Test (DCT)

For some integer N_0, suppose $0 \leq a_n \leq b_n$ for all $n \geq N_0$.

- If $\sum b_n$ converges, then $\sum a_n$ converges.
- If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Limit Comparison Test (LCT)

Suppose $a_n \geq 0$ and $b_n > 0$. Let $L = \lim_{n\to\infty} \frac{a_n}{b_n}$.

- If $0 < L < \infty$: $\sum a_n$ and $\sum b_n$ have the same convergence behaviour.
- If $L = 0$ and $\sum b_n$ converges: $\sum a_n$ converges.
- If $L = \infty$ and $\sum b_n$ diverges: $\sum a_n$ diverges.

Direct Comparison Test — WW10 P14 (4)

Does $\displaystyle\sum_{n=1}^{\infty} \frac{n^3 + 9n^2 + 6n + 1}{4n^5 + 8n^3 - 100n}$ converge?

Solution. For large n: dominant terms give $\frac{n^3}{4n^5} = \frac{1}{4n^2}$. Since $n^3+9n^2+6n+1 \leq n^3$ and $4n^5 + 8n^3 - 100n \geq n^5$ for all $n \geq N_0$:

$$0 \leq \frac{n^3 + 9n^2 + 6n + 1}{4n^5 + 8n^3 - 100n} \leq \frac{n^3}{n^5} = \frac{1}{n^2}.$$

Since $\sum 1/n^2$ converges ($p = 2 > 1$), by DCT the series **converges**.

Limit Comparison Test — WW10 P17 (1)

Does $\displaystyle\sum_{n=1}^{\infty} \frac{3n^2+5}{9n^3+n^2+10n+100}$ converge or diverge?

Solution. Dominant terms: $3n^2/9n^3 = 1/(3n)$. Choose $b_n = 1/n$.

Check signs: $a_n > 0$, $b_n > 0$ for all $n \geq 1$.✓

Compute L:

$$L = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{3n^3+5n}{9n^3+n^2+10n+100} = \frac{3}{9} = \frac{1}{3}.$$

Since $0 < L = \frac{1}{3} < \infty$ and $\sum 1/n$ **diverges** ($p = 1$), by LCT the series **diverges**.

Tips & Tricks – Comparison Tests for Series

- **Choosing a comparison series:** keep only the dominant term of numerator and denominator as $n \to \infty$ — ignore constants, lower-order terms, and $\pm$ tweaks.
- **DCT direction matters:**
  - To prove *convergence*: need $0 \leq a_n \leq b_n$ and $\sum b_n$ converges.
  - To prove *divergence*: need $0 \leq b_n \leq a_n$ and $\sum b_n$ diverges.
- **LCT is usually easier**: if $L = \lim a_n/b_n$ exists and $0 < L < \infty$, both series behave the same — no inequality needed.
- **Standard reference series**: p-series $\sum 1/n^p$ and geometric $\sum r^n$ cover almost all cases.
- On an exam, always write the full argument: (1) define b_n, (2) check signs, (3) compute limit or establish inequality, (4) state convergence of $\sum b_n$ with reason, (5) conclude by the test name.

3.4 Week 10 Small Class – Alternating Series; Absolute Convergence (SC10)

Alternating Series Test (AST)

If $a_n > 0$, $\{a_n\}$ is monotonically decreasing, and $a_n \to 0$, then $\displaystyle\sum_{n=1}^{\infty}(-1)^n a_n$ **converges**.

Error bound: $|S - S_N| \leq a_{N+1}$, where S_N is the N-th partial sum.

Absolute and Conditional Convergence

$\sum a_n$ **converges absolutely** if $\sum |a_n|$ converges. Absolute convergence implies convergence.

$\sum a_n$ **converges conditionally** if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Alternating Series Test and partial-sum bound — WW10 P8

Consider $S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$.

(a) Show the series converges. (b) Find an interval of length ≤ 0.05 containing S.

(a) Let $a_n = 1/n^2$: positive, strictly decreasing, $a_n \to 0$. By the AST, the series **converges**.

(b) The AST guarantees S lies between consecutive partial sums. Compute:

$$S_3 = 1 - \tfrac{1}{4} + \tfrac{1}{9} = \tfrac{49}{36}, \qquad S_4 = S_3 - \tfrac{1}{16} = \tfrac{745}{576}.$$

The error after N terms satisfies $|S - S_N| \leq a_{N+1}$. For $a_5 = \frac{1}{25} < \frac{1}{20} = 0.05$:

$$S \in \left[\tfrac{745}{576}, \tfrac{49}{36}\right].$$

Absolute convergence via comparison

Does $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$ converge?

Solution. $|\sin n/n^2| \leq 1/n^2$; $\sum 1/n^2$ converges. By DCT, $\sum |\sin n/n^2|$ converges. Therefore $\sum \sin n/n^2$ converges absolutely (and hence converges).

Tips & Tricks – Alternating Series & Absolute Convergence

- **AST conditions:** the series $\sum(-1)^n a_n$ converges if (1) $a_n > 0$, (2) a_n is eventually decreasing, and (3) $a_n \to 0$. All three are required.
- **Proving Condition 2:** set $a_n - a_{n+1}$, do some algebra and prove this equation is larger than or equal to zero. Or remember $x > y \implies 1/x < 1/y$.
- **Error bound for AST:** $|S - S_N| \leq a_{N+1}$. To guarantee accuracy within ε, find the smallest N with $a_{N+1} \leq \varepsilon$.
- **Absolute convergence** ($\sum |a_n|$ converges) $\Rightarrow$ convergence of $\sum a_n$. This is one way only!
- **Conditionally convergent:** converges, but not absolutely (e.g. $\sum(-1)^{n+1}/n$). These are more delicate — rearranging terms can change the sum!
- Quick strategy: test $\sum |a_n|$ first. If it converges, you're done (absolutely convergent). If not, try the AST on the original alternating series.

3.5 Week 11 – Ratio Test; Power Series (LC11)

Ratio Test

Given $\sum a_n$, let $r = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$.

- $r < 1$: the series **converges** (absolutely).
- $r > 1$ (or $r = +\infty$): the series **diverges**.
- $r = 1$ or the limit does not exist: **inconclusive** — use another test.

Power Series and Radius of Convergence

A **power series** centred at c has the form $\sum_{n=0}^{\infty} A_n(x-c)^n$.

For every power series there exists a **radius of convergence** $R \in [0, +\infty]$ such that:

- The series converges (absolutely) for all x with $|x - c| < R$.
- The series diverges for all x with $|x - c| > R$.
- The endpoints $x = c \pm R$ must be checked separately.

To find R via the Ratio Test: $\frac{1}{R} = \lim_{n\to\infty} \left|\frac{A_{n+1}}{A_n}\right|$.

Ratio Test with factorials — WW11 P9

Does $\sum_{n=1}^{\infty} \frac{n^n}{6^n \cdot n!}$ converge or diverge?

Solution. Let $a_n = \frac{n^n}{6^n \cdot n!}$. Apply the Ratio Test:

$$r = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{(n+1)^{n+1}}{6^{n+1}(n+1)!} \cdot \frac{6^n n!}{n^n} = \lim_{n\to\infty} \frac{(n+1)^n}{6\,n^n} = \frac{1}{6} \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = \frac{e}{6} < 1.$$

Since $r = e/6 < 1$, the series **converges** by the Ratio Test.
Key step: $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$.

Radius of convergence — WW11 P20

Find the radius of convergence of $\displaystyle\sum_{n=1}^{\infty} \frac{(5x-4)^n}{n^2}$.

Solution. Let $a_n = (5x-4)^n/n^2$. Apply the Ratio Test:

$$r = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} |5x-4| \cdot \frac{n^2}{(n+1)^2} = |5x-4|.$$

The series converges when $|5x-4| < 1$, i.e. $\left|x - \frac{4}{5}\right| < \frac{1}{5}$.

Hence the **radius of convergence** is $R = \frac{1}{5}$, centred at $x = \frac{4}{5}$.

Interval of convergence for a geometric series — WW11 P27a

Find the set of real numbers t for which $A(t) = \displaystyle\sum_{n=5}^{\infty} (9-18t)^n$ converges.

Solution. This is a geometric series with ratio $r = 9 - 18t$. It converges iff $|r| < 1$:

$$|9-18t| < 1 \iff -1 < 9-18t < 1 \iff 8 < 18t < 10 \iff \frac{4}{9} < t < \frac{5}{9}.$$

The set of convergence is $t \in \left(\frac{4}{9}, \frac{5}{9}\right)$.

Tips & Tricks – Ratio Test & Power Series

Ratio Test:

- Best for series with **factorials** ($n!$) or **exponentials** (a^n) — these cancel beautifully in $|a_{n+1}/a_n|$.
- If $r = 1$, the Ratio Test is *inconclusive* — switch to the p-test, DCT, LCT, or AST.
- Polynomial factors like n^k always give ratio $\to 1$ alone, so pair them with an exponential or factorial for the Ratio Test to work.

Power Series & Radius of Convergence:

- Apply the Ratio Test to $\sum A_n(x-c)^n$: solve $\lim |A_{n+1}/A_n| \cdot |x-c| < 1$ to get the interval of convergence $(c-R, c+R)$.
- Endpoints $x = c \pm R$ must be checked *separately* (the Ratio Test is inconclusive there).
- Operations on power series **preserve** R (or can only increase it): term-by-term differentiation and integration are valid inside the interval of convergence.
- **Substitution trick:** replace x by $g(x)$ in a known series. E.g. substitute $-x^2$ into e^x to get e^{-x^2}.

3.6 Week 11 Small Class – Power Series and Taylor Series (SC11)

Manipulating Known Power Series

Starting from the geometric series $\frac{1}{1-y} = \sum_{n=0}^{\infty} y^n$, $|y| < 1$, we can build new power series by:

- **Substitution:** replace y by a function $g(x)$ to get a series in x. Adjust the radius of convergence by solving $|g(x)| < 1$.
- **Multiplication by x^k:** multiply each term by x^k (does not shrink R).
- **Term-by-term integration/differentiation:** valid inside $(c - R, c + R)$; R does not decrease.

Radius of convergence — WW11 P22

Find the radius of convergence of $\sum_{n=1}^{\infty} (-1)^n n\, 2^n x^n$.

Solution. Let $a_n = (-1)^n n\, 2^n x^n$. Apply the Ratio Test:

$$r = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} 2|x| \cdot \frac{n+1}{n} = 2|x|.$$

The series converges when $2|x| < 1$, i.e. $|x| < \frac{1}{2}$.

The **radius of convergence** is $R = \frac{1}{2}$.

Taylor Series and Maclaurin Series

If f has derivatives of all orders at $x = a$, the **Taylor series** of f centred at a is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

When $a = 0$ this is called the **Maclaurin series** of f.

A Taylor series is the limit $n \to \infty$ of the nth Taylor polynomial $T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$, and is itself a power series (with some radius of convergence R).

Power series via differentiation — WW12 P7

Find the power series centred at $x = 0$ for $f(x) = \dfrac{5}{(1-3x)^2}$ and its radius of convergence.

Solution. Start from the geometric series:

$$\frac{1}{1-3x} = \sum_{n=0}^{\infty}(3x)^n = \sum_{n=0}^{\infty} 3^n x^n, \quad |x| < \tfrac{1}{3}.$$

Differentiate both sides with respect to x:

$$\frac{3}{(1-3x)^2} = \sum_{n=1}^{\infty} n \cdot 3^n x^{n-1}.$$

Divide by 3, then multiply by 5:

$$\frac{5}{(1-3x)^2} = 5\sum_{n=0}^{\infty}(n+1)\cdot 3^n x^n = 5 + 30x + 135x^2 + 540x^3 + \cdots$$

The **radius of convergence** is $R = \dfrac{1}{3}$ (unchanged from the geometric series).

Power series for an integral — WW12 P4

Find the power series centred at $x = 0$ for $f(x) = \displaystyle\int_0^x \frac{\arctan(8t)}{t}\,dt$, and its radius of convergence.

Solution.

Step 1. Use $\arctan u = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n u^{2n+1}}{2n+1}$ with $u = 8t$:

$$\arctan(8t) = \sum_{n=0}^{\infty} \frac{(-1)^n 8^{2n+1} t^{2n+1}}{2n+1}.$$

Step 2. Divide by t:

$$\frac{\arctan(8t)}{t} = \sum_{n=0}^{\infty} \frac{(-1)^n 8^{2n+1} t^{2n}}{2n+1}.$$

Step 3. Integrate term-by-term from 0 to x:

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n 8^{2n+1}}{(2n+1)^2} x^{2n+1} = 8x - \frac{8^3}{9}x^3 + \frac{8^5}{25}x^5 - \cdots$$

Radius of convergence: The arctan series requires $|8t| \le 1$, giving $R = \dfrac{1}{8}$.

Taylor polynomial and a limit — WW12 P11

(a) Find the Taylor polynomial of degree 3 for $\sin(x)$ about $x = 0$.
(b) Use it to approximate $\frac{\sin x}{x}$ and evaluate $\lim_{x\to 0} \frac{\sin x}{x}$.

Solution.
(a) The Maclaurin series for $\sin x$ starts $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$, so:

$$P_3(x) = x - \frac{x^3}{6}.$$

(b) Dividing by x:

$$\frac{\sin x}{x} \approx \frac{P_3(x)}{x} = 1 - \frac{x^2}{6}.$$

Taking the limit:

$$\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \left(1 - \frac{x^2}{6}\right) = 1.$$

Tips & Tricks – Power Series & Taylor Series (SC11)

- **Two methods to find a Taylor series:** (1) extend the Taylor polynomial using the formula $\sum f^{(n)}(a)/n!\,(x-a)^n$; (2) manipulate a known power series (substitution, multiply, differentiate, integrate).
- **Substitution adjusts R:** if the geometric series converges for $|y| < 1$ and you set $y = g(x)$, solve $|g(x)| < 1$ to find the new interval/radius of convergence.
- **Term-by-term integration/differentiation:** valid inside $(c-R, c+R)$ and does not shrink R. Always find $+C$ by plugging in a known value (e.g. $x = 0$).
- **Approximating π:** since $\arctan(1) = \pi/4$, the series for arctan at $x = 1$ gives $\pi = 4\sum_{n=0}^{\infty}(-1)^n/(2n+1)$, a slowly converging alternating series.
- **Stick with $|x| < R$:** in this course the focus is on finding the *radius* of convergence. When working through a convergence inequality (e.g. from the Ratio Test), keep the expression in the form $|x| < R$ throughout — do *not* expand it into $-R < x < R$ prematurely. Your final goal is to identify R and state "the radius of convergence is R".

3.7 Week 12 – Taylor Series (LC12)

Taylor coefficients via known series — WW12 P3

Find the first six coefficients $c_0, \ldots, c_5$ in the power series $8x^2e^{-7x} = \sum_{n=0}^{\infty} c_n x^n$.

Solution. Substitute $u = -7x$ into $e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!}$:

$$e^{-7x} = \sum_{n=0}^{\infty} \frac{(-7)^n}{n!} x^n.$$

Multiply by $8x^2$ (shifting the index by 2):

$$8x^2e^{-7x} = 8\sum_{n=0}^{\infty} \frac{(-7)^n}{n!} x^{n+2} = 8\sum_{n=2}^{\infty} \frac{(-7)^{n-2}}{(n-2)!} x^n.$$

Since there are no constant or linear terms: $c_0 = c_1 = 0$. Reading off the remaining coefficients:

$$c_2 = 8, \quad c_3 = -56, \quad c_4 = 196, \quad c_5 \approx -457.3.$$

Limit via Taylor series — WW12 P12

Evaluate $\lim_{x\to 0} \frac{\log(1-x) + x + \frac{x^2}{2}}{4x^3}$.

Solution. Use $\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots$ (valid for $|x| < 1$). Substituting:

$$\log(1-x) + x + \frac{x^2}{2} = \left(-x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots\right) + x + \frac{x^2}{2} = -\frac{x^3}{3} + O(x^4).$$

Therefore:

$$\lim_{x\to 0} \frac{\log(1-x) + x + x^2/2}{4x^3} = \lim_{x\to 0} \frac{-x^3/3 + O(x^4)}{4x^3} = \frac{-1/3}{4} = -\frac{1}{12}.$$

Series representation of an integral — WW12 P5

Find the power series centred at $x = 0$ for $f(x) = \int_0^x \frac{\sin(3t)}{2t}\,dt$, and its radius of convergence.

Solution. Use $\sin(u) = \sum_{n=0}^{\infty} \frac{(-1)^n u^{2n+1}}{(2n+1)!}$ with $u = 3t$:

$$\frac{\sin(3t)}{2t} = \frac{1}{2t}\sum_{n=0}^{\infty} \frac{(-1)^n (3t)^{2n+1}}{(2n+1)!} = \frac{1}{2}\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1}}{(2n+1)!}\, t^{2n}.$$

Integrate term-by-term from 0 to x:

$$f(x) = \frac{1}{2}\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1}}{(2n+1)!\,(2n+1)}\, x^{2n+1}.$$

The first three nonzero terms: $f(x) = \frac{3}{2}x - \frac{3}{4}x^3 + \frac{81}{400}x^5 - \cdots$

Radius of convergence: The sin series converges for all u, so $R = \infty$.

Summing a series using Taylor series — LC12 Final Exam Q8 style

Evaluate $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \cdot 2^n}$.

Solution. Let $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$. Note $f(0) = 0$ and

$$f'(x) = \sum_{n=1}^{\infty} (-1)^{n-1} x^{n-1} = \sum_{k=0}^{\infty} (-x)^k = \frac{1}{1+x}, \quad |x| < 1.$$

Integrating: $f(x) = \ln(1+x)$. Substituting $x = \frac{1}{2}$:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \cdot 2^n} = \ln\left(\frac{3}{2}\right).$$

Tips & Tricks – Taylor Series

Building new series from known ones (faster than computing derivatives):
- **Substitute** $g(x)$ for x: e.g. replace x with $-x^2$ in $e^x = \sum x^n/n!$ to get e^{-x^2}.
- **Multiply by** x^k: shifts all powers up by k.
- **Differentiate term-by-term**: gives the series for $f'(x)$, valid on the same interval.
- **Integrate term-by-term**: gives the series for $\int f(x)\,dx$; add $+C$ and use an initial condition to fix it.

Extracting Taylor coefficients:
- Match powers: coefficient of x^k in the series is $\dfrac{f^{(k)}(0)}{k!}$, so $f^{(k)}(0) = k! \times$ (coefficient of x^k).
- If a power x^k is *missing* from the series, then $f^{(k)}(0) = 0$.

Limits using Taylor series (instead of L'Hôpital):
- Expand numerator and denominator to the *lowest non-cancelling power*. Cancel common factors of x^k, then take the limit by setting $x = 0$.
- This is usually faster than repeated L'Hôpital's rule for 0/0 forms.

Standard series to memorise (also provided on the final exam):

$$\begin{aligned}
&\frac{1}{1-x} = \textstyle\sum_{n=0}^{\infty} x^n && |x| < 1\\
&e^x = \textstyle\sum_{n=0}^{\infty} \dfrac{x^n}{n!} && \text{all } x\\
&\sin x = \textstyle\sum_{n=0}^{\infty} \dfrac{(-1)^n x^{2n+1}}{(2n+1)!} && \text{all } x\\
&\cos x = \textstyle\sum_{n=0}^{\infty} \dfrac{(-1)^n x^{2n}}{(2n)!} && \text{all } x\\
&\ln(1+x) = \textstyle\sum_{n=1}^{\infty} \dfrac{(-1)^{n-1}}{n} x^n && -1 < x \le 1\\
&\arctan x = \textstyle\sum_{n=0}^{\infty} \dfrac{(-1)^n}{2n+1} x^{2n+1} && |x| \le 1
\end{aligned}$$

3.8 Week 12 Small Class – Taylor Truncation Error (SC12)

Lagrange Remainder (Error Bound)

If f is $(n+1)$ times differentiable and $T_n(x)$ is the nth Taylor polynomial of f about $x = a$, define the truncation error $E_n(x) = f(x) - T_n(x)$. The **Lagrange Remainder Formula** gives:

$$|E_n(x)| \le \frac{M_{n+1}}{(n+1)!}\,|x-a|^{n+1},$$

where M_{n+1} is any constant satisfying $|f^{(n+1)}(c)| \le M_{n+1}$ for all c between a and x.

Key insight: the bound equals "an upper bound on the next term in the series" — the exact $(n+1)$th derivative replaced by its worst-case value M_{n+1}.

Approximating e and bounding the error — SC12 Q2–3

Use the Taylor polynomial $T_4(x)$ for $f(x) = e^x$ centred at $a = 0$ to approximate e, then bound the error.

Step 1 — Approximate $e = f(1)$:

$$T_4(1) = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} = \frac{65}{24} \approx 2.7083.$$

Step 2 — Apply Lagrange with $n = 4$, $x = 1$, $a = 0$:
All derivatives of e^x equal e^x, so we need $M_5 \geq e^c$ for all $c \in (0, 1)$. Since e^c is increasing, its max is $e^1 = e < 3$, so take $M_5 = 3$:

$$|E_4(1)| \leq \frac{3}{5!} \cdot |1 - 0|^5 = \frac{3}{120} = \frac{1}{40} = 0.025.$$

So $T_4(1) = 65/24$ is guaranteed correct to **1 decimal place**.

Note: We use 3 rather than e because we are trying to *approximate* e — it is not yet a known number. Any value $> e$ works; a smaller bound gives a tighter estimate.

Lagrange error bound for $\sqrt{x+1}$ — WW13 P2

Let $f(x) = \sqrt{x+1}$. Compute $T_1(10)$ and $T_2(10)$ for f expanded about $a = 8$, and bound $|E_2(10)|$.

Solution. Compute derivatives at $a = 8$:

$$f(8) = 3, \quad f'(x) = \tfrac{1}{2}(x+1)^{-1/2} \Rightarrow f'(8) = \tfrac{1}{6}, \quad f''(x) = -\tfrac{1}{4}(x+1)^{-3/2} \Rightarrow f''(8) = -\tfrac{1}{108}.$$

$$T_1(10) = 3 + \tfrac{1}{6} \cdot 2 = \tfrac{10}{3} \approx 3.333, \qquad T_2(10) = \tfrac{10}{3} + \frac{-1/108}{2} \cdot 4 \approx 3.315.$$

Error bound ($n = 2$): $f'''(x) = \frac{3}{8}(x+1)^{-5/2}$ is decreasing on $[8, 10]$, so $|f'''(t)| \leq f'''(8) = \frac{3}{8 \cdot 243}$ for all $t \in [8, 10]$. Applying Lagrange:

$$|E_2(10)| \leq \frac{3/(8 \cdot 243)}{3!} \cdot (10 - 8)^3 = \frac{3 \cdot 8}{8 \cdot 243 \cdot 6} = \frac{1}{486} \approx 0.0021.$$

Since $f''(x) < 0$ and $f'''(x) > 0$, T_2 **underestimates** $f(10) = \sqrt{11}$.

Tips & Tricks – Taylor Truncation Error (SC12)

- **Error ≈ next term:** The bound $\frac{M_{n+1}}{(n+1)!}|x-a|^{n+1}$ is the $(n+1)$th series term with the exact derivative replaced by its max M_{n+1}.
- **Finding M_{n+1}:** Bound $|f^{(n+1)}(t)|$ over the interval $[a, x]$. For monotone derivatives, just evaluate at the endpoint giving the largest value. Use known inequalities (e.g. $e < 3$) when approximating transcendental constants.
- **Over- vs. underestimate:** determined by the sign of $f^{(n+1)}(t)$ on the interval. Positive ⇒ underestimate; negative ⇒ overestimate.
- **Decimal places:** if $|E_n(x)| \leq 5 \times 10^{-k}$, the approximation is correct to $k-1$ decimal places.
- **More terms ⇒ smaller error:** $|x-a|^{n+1}/(n+1)!$ shrinks rapidly (factorial beats power).

A Few Words Before You Go

Dear all,
As we wrap up our time together in this calculus class, I want to leave you with a few thoughts—not just about math, but about what lies ahead for you.
Calculus is about change, and you're stepping into one of the most important periods of change in your lives. Your undergraduate years will go by faster than you expect, but they will also shape you in ways you can't fully see yet.
From my own experience, some of the most meaningful things in life happen during this time. I met some truly amazing people during my undergraduate years—people who are still close to me today. I even met my life-long partner back then, something I never could have planned.
Later on, when I was doing my internship, I had the chance to work with people who were incredibly talented and deeply passionate about what they do. That experience taught me something important: no matter how much you achieve, always stay humble. There will always be people who know more, who think differently, and who can inspire you to grow.
You are also entering a world that is changing rapidly with AI and new technologies. Things will keep evolving, and that can feel uncertain—but it also means there are so many possibilities ahead of you.
So stay curious, stay open, and don't be afraid to fail. Failure is not the opposite of success—it's part of it. Just like in calculus, understanding often comes after trying, struggling, and trying again.
Most importantly, I hope you have a wonderful undergraduate life. Learn deeply, build meaningful friendships, and take care of yourselves along the way.
Thank you for being part of this class—it's been a real pleasure teaching you.

Take care, and all the best for what's ahead.